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RESEARCH ARTICLE

Widespread variation in functional trait–vital rate relationships in tropical tree seedlings across a precipitation and soil phosphorus gradient

Luke Browne[1] | Lars Markesteijn[2,3,4] | Eric Manzané-Pinzón[2] | S. Joseph Wright[2] | Robert Bagchi[5] | Bettina M. J. Engelbrecht[2,6] | F. Andrew Jones[2,7] | Liza S. Comita[1,2]

[1]School of the Environment, Yale University, New Haven, Connecticut, USA; [2]Smithsonian Tropical Research Institute, Balboa, Panama; [3]Departamento de Biología y Geología, Física y Química inorgánica. ESCET, Universidad Rey Juan Carlos, Madrid, Spain; [4]School of Natural Sciences, Bangor University, Bangor, UK; [5]Department of Ecology and Evolutionary Biology, University of Connecticut, Storrs, Connecticut, USA; [6]Department of Plant Ecology, Bayreuth Center of Ecology and Environmental Research (BayCEER), University of Bayreuth, Bayreuth, Germany and [7]Department of Botany and Plant Pathology, Oregon State University, Corvallis, Oregon, USA

Correspondence
Luke Browne
Email: lukembrowne@gmail.com

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Abstract

1. A fundamental assumption of functional ecology is that functional traits are related to interspecific variation in performance. However, the relationship between functional traits and performance is often weak or uncertain, especially for plants. A potential explanation for this inconsistency is that the relationship between functional traits and vital rates (e.g. growth and mortality) is dependent on local environmental conditions, which would lead to variation in trait–rate relationships across environmental gradients.
2. In this study, we examined trait–rate relationships for six functional traits (seed mass, wood density, maximum height, leaf mass per area, leaf area and leaf dry matter content) using long-term data on seedling growth and survival of woody plant species from eight forest sites spanning a pronounced precipitation and soil phosphorus gradient in central Panama.
3. For all traits considered except for leaf mass per area–mortality, leaf mass per area–growth and leaf area–mortality relationships, we found widespread variation in the strength of trait–rate relationships across sites. For some traits, trait–rate relationships showed no overall trend but displayed wide site-to-site variation. In a small subset of cases, variations in trait–rate relationships were explained by soil phosphorus availability.
4. Our results demonstrate that environmental gradients have the potential to influence how functional traits are related to growth and mortality rates, although much variation remains to be explained. Accounting for site-to-site variation may help resolve a fundamental issue in trait-based ecology—that traits are often weakly related to performance—and improve the utility of functional traits for explaining key ecological and evolutionary processes.

KEYWORDS
forest dynamics, Panama, rainfall gradient, soil nutrients, tropics

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1 | INTRODUCTION

Trait-based ecology assumes that functional traits influence fitness via their effects on vital rates including recruitment, growth and survival (Violle et al., 2007). These functional trait–vital rate relationships (hereafter trait–rate relationships) provide the basis for understanding and predicting fundamental ecological and evolutionary processes such as species coexistence, community assembly, response to environmental change and ecosystem functioning (Cadotte et al., 2011; Kraft et al., 2015; Lavorel & Garnier, 2002; McGill et al., 2006). While a number of studies have found some consistency in trait–rate relationships (Poorter et al., 2008; Visser et al., 2016; Wright et al., 2010), the relationship between functional traits and vital rates in plants is often weak or non-existent (Paine et al., 2015; Poorter et al., 2018; Worthy & Swenson, 2019), calling into question one of the foundational assumptions of trait-based ecology and limiting the utility of trait-based approaches (Swenson et al., 2020; Yang et al., 2018). A potential explanation for this inconsistency is that phenotypes and vital rates interact with local environmental conditions. This in turn would lead to variation in trait–rate relationships across environmental gradients that would obscure and weaken general patterns within and among plant species (Li et al., 2022; Swenson et al., 2020; Yang et al., 2018). Thus, resolving the degree to which trait–rate relationships vary across environmental gradients will likely help improve the use of functional traits as a paradigm to explain ecological and evolutionary processes (Laughlin, 2018; Yang et al., 2018).

Because of trade-offs in resource allocation, only a subset of all possible life-history strategies are likely to provide fitness advantages in a given environment (Stearns, 1992). This variation in life-history strategies among species can potentially be correlated with a relatively small number of functional traits, which is especially useful in species-diverse tropical tree communities, where detailed information on demography and life history is typically lacking for most species (Díaz et al., 2016; Laughlin, 2014; Worthy & Swenson, 2019). For example, species with high wood density, high leaf mass per unit area and high leaf dry matter content tend to have a more resource-conservative strategy along the fast-slow life-history continuum that prioritizes defence and storage, with greater investment in the physical protection of leaves, lower respiration rates, greater stress tolerance and a longer leaf life span (Alvarez-Clare & Kitajima, 2007; Chave et al., 2009; Poorter, 2009; Poorter & Rozendaal, 2008; Reich, 2014; Weiher et al., 1999; Wright et al., 2004; Wright & Westoby, 2002). Trees with higher maximum height and larger leaves tend to have above average growth and survival rates and lower recruitment rates for individuals >1 cm DBH (diameter at breast height), but lower seedling survival and growth (Kohyama et al., 2003; Rüger et al., 2018, 2020), representing a trade-off between growth and survival with tree stature, known as the 'stature-recruitment' axis. Seed mass is positively correlated with seedling recruitment rates and greater stress tolerance at early life stages (Moles & Westoby, 2006; Muller-Landau, 2010). Despite the wealth of research into the general relationships between functional traits, life-history strategies and plant performance, the role of environmental context in driving these relationships remains underexplored (Ackerly, 2003; Laughlin, 2018; Laughlin & Messier, 2015).

Resource availability within and across environments may be an important factor influencing trait–rate relationships. For example, in high-resource environments, traits that maximize carbon gain are advantageous and can result in relatively high growth rates, whereas the same strategy can be disadvantageous in low-resource environments and result in relatively low growth rates, due to trade-offs between growth and stress resistance (Figure 1a) (Coley et al., 1985; Grime, 1977; Kobe, 1999; Reich, 2014; Russo et al., 2005). Across a resource availability gradient, the strength of trait–growth rate relationships may then show a predictable pattern where correlations between traits and growth rates are weakest in low-resource environments and strongest in high-resource environments (Figure 1a). Conversely, for mortality rates, trait–mortality correlations would be strongest in low-resource environments and weakest in high-resource environments (Figure 1b). Evidence of these relationships have been found within a Bornean rain forest, where Russo et al. (2007) demonstrated that resource-acquisitive species suffered higher mortality in low-resource soils compared with more fertile soils. Variation in light availability also leads to similar interactions between vital rates and local environment (Bloor & Grubb, 2003; Kobe, 1999; Walters & Reich, 1996; Wright et al., 2010). While recent studies have found in some cases that explicitly accounting for trait by environment interactions improves models of plant performance (Jiang & Jin, 2021; Laughlin et al., 2018; Li et al., 2022; Worthy et al., 2020; Yang et al., 2021), resolving the degree to which local environment influences trait–rate relationships is a key research priority for the field of functional ecology to understand the contexts in which functional traits are strong or weak predictors of fitness.

In this study, we tested the hypothesis that trait–rate relationships vary with local environmental context. We combined functional trait data with long-term monitoring of growth and mortality rates of tropical seedling communities across a strong precipitation and soil nutrient gradient in central Panama. The severity of the annual dry season varies widely across the Isthmus of Panama (Figure 2), with forests on the Caribbean side experiencing less severe dry seasons than forests on the Pacific side (Condit et al., 2013). Additionally, there is a strong variation in soil nutrients, particularly soil phosphorus availability (Condit et al., 2013). Both dry season severity (DSS) and soil phosphorus availability are important predictors of seedling performance and species distributions across central Panama and other tropical regions (Alvarez-Clare et al., 2013; Condit et al., 2013; Gaviria et al., 2017; Wright et al., 2011; Zalamea et al., 2016). We focused on seedling communities because the understory dynamics at these early life stages influence future patterns of forest structure and diversity (Green et al., 2014; Poorter, 2007) and relatively little is known about trait–rate relationships in tropical seedling communities (but see Umaña et al., 2017). Specifically,

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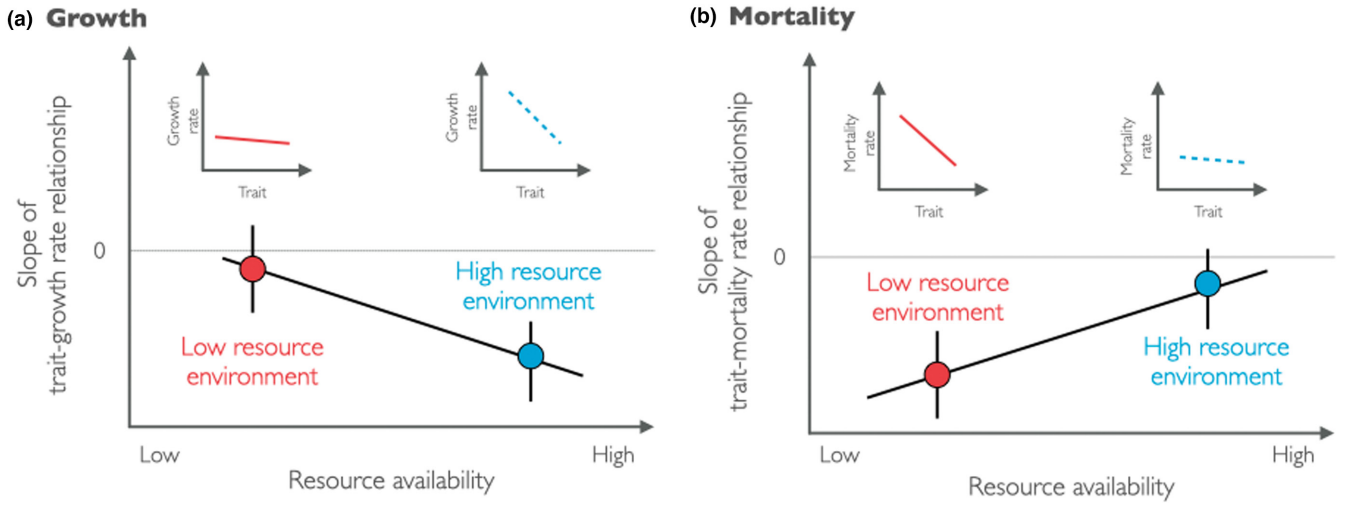


FIGURE 1 Conceptual diagram showing how (a) growth rates and (b) mortality rates may vary based on trait values and how these relationships may interact depending on resource availability. Both panels show expected relationships for a trait assumed to be negatively related to growth and mortality. For traits positively related to growth and mortality, we would expect opposite relationships (not shown). Insets show the relationships between growth and mortality rates and a hypothetical trait in both low-resource (red solid line) and high-resource (blue dashed line) environments. The corresponding slopes of these lines are displayed on the larger plot, with hypothetical 95% credible intervals. The solid black line shows the hypothetical relationship between the slope of trait–rate relationships going from low- to high-resource environments. The dashed line shows where the trait–rate relationship is 0.

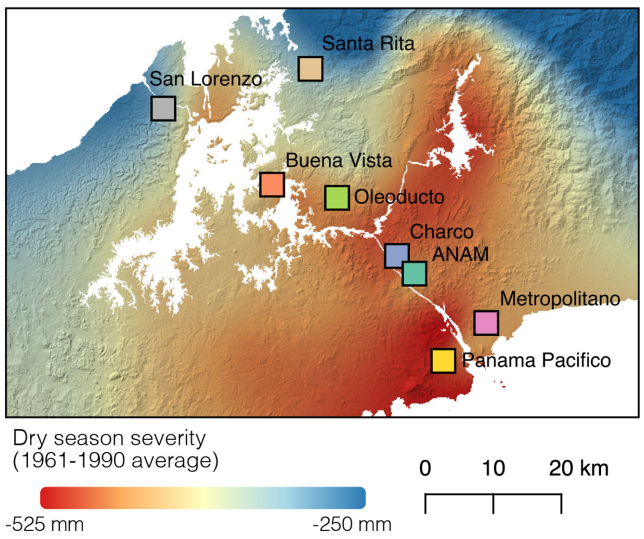


FIGURE 2 Map of study area showing the eight sites across the isthmus of Panama containing seedling plots, along with shading indicating dry season severity, with redder shades showing more intense dry seasons compared with bluer shades. Colour coding of sites corresponds to legends in Figures 4 and 5.

we asked the following questions: (1) Across all sites, do we observe relationships between functional traits and growth and mortality rates (Table 1)? (2) Does the strength of the trait–rate relationships vary among sites? (3) If so, is that variation predicted by local DSS and/or soil phosphorus availability (Table 1)?

Based on previous studies in both adults and seedlings (Alvarez-Clare & Kitajima, 2007; Chave et al., 2009; Kohyama et al., 2003; Moles & Westoby, 2006; Muller-Landau, 2010; Poorter, 2009; Poorter & Rozendaal, 2008; Reich, 2014; Rüger et al., 2018, 2020; Weiher et al., 1999; Wright et al., 2004; Wright & Westoby, 2002), we predicted that growth rates would be negatively related to wood density, leaf mass per unit area, leaf dry matter content, maximum height, seed mass and leaf area and that the strength of this negative relationship would be stronger in high-resource environments compared with low-resource environments (Table 1). We predicted that mortality rates would be negatively related to wood density, leaf mass per unit area, leaf dry matter content and seed mass, with the strength of this relationship being weaker in high-resource compared with low-resource environments (Table 1). Finally, we predicted that mortality rates would be positively related to maximum height and leaf area, with the strength of this relationship increasing in high-resource environments compared with low-resource environments (Table 1).

2 | MATERIALS AND METHODS

2.1 | Study area and seedling censuses

This study was conducted in eight 1-ha plots in seasonally moist tropical forests across the Isthmus of Panama (Figure 2). These plots span a relatively short 65-km gradient, where mean annual precipitation ranges from ~3200 to ~1600 mm. Plant-available soil phosphorus levels (hereafter soil phosphorus levels) also vary strongly among sites and range from 3.0 to 22.8 mg/kg (Condit et al., 2013). Within each 1-ha plot, 400 1-m^2 seedling plots were established

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TABLE 1 Expected predictions of directionality of growth and mortality rates in seedlings and six functional traits. Shown are both the overall relationship (i.e. averaged across sites) and how the strength of the relationship is expected to change going form low- to high-resource environments. '−' indicates an expected negative relationship and '+' indicates an expected positive relationship. A representative graphical depiction of these relationships is shown in Figure 1.

	Growth		Mortality	
Trait	Overall relationship	Change from low to high resources	Overall relationship	Change from low to high resources
Wood density	−	−	−	+
LMA	−	−	−	+
Leaf dry matter content	−	−	−	+
Max height	−	−	+	−
Seed mass	−	−	−	+
Leaf area	−	−	+	−

from September–December 2013, where within each 1-m^2 plot, all woody seedlings ≥200 mm in height and <1 cm DBH (diameter at 1.3 m above ground) were tagged, identified and measured following Comita et al. (2007). Seedling plots were re-censused annually near the beginning of the annual dry season (November–February). The maximum stem height of all marked seedlings was measured, and all seedlings were evaluated whether they were alive or dead. Any new seedlings that recruited into the size criterion (≥200 mm height) were entered into the census. Seedlings without a definitive species identification (n = 519 individuals) and lianas were excluded from analyses. Due to limited access, one site, Oleoducto, in 2019 or 2020, and another site, Panamá Pacifico, in 2018, were not censused. The last census included in this study occurred in 2021, for a total of seven annual census intervals. The overall dataset, before filtering based on trait availability, included 28,303 observations of 9267 individuals belonging to 358 tree species. The research was conducted in Panama under permits from the Ministry of Environment (MiAmbiente) and the Agencia Panama Pacífico (APP).

2.2 | Trait data

We focused on six widely available, species-level traits from Wright et al. (2010) that are related to both the fast-slow and stature-recruitment axes of life-history variation: wood density (g/cm^3, 247 species), LMA (leaf mass per unit area, g/m^2, 184 species), leaf dry matter content (g/g, 184 species), maximum height (m, 186 species), seed mass (g, 180 species) and leaf area (cm^2, 184 species). A full description of the methods of trait data collection are available in Wright et al. (2010). Briefly, wood density data was collected from ~5 adult individuals per species within 15 km of the Barro Colorado Island (BCI) 50-ha forest dynamics plot, which is located near the center of the rainfall gradient. Leaf traits were collected from leaves receiving indirect sunlight from six of the smallest individuals of each species at the BCI 50-ha plot. Maximum height was estimated as the mean height of the six individuals with the largest DBH in the 50-ha plot on BCI and a nearby 38.4 ha plot (King et al., 2006; Wright et al., 2010). Seed mass is the mean dry mass that includes the endosperm and embryo only, measured from 1 to 11 individuals and 1 to 139 seeds per species. Pairwise correlations among traits ranged from Pearson's r = 0.01–0.57 (Figure S1). It is also important to note that the trait data used in this study was obtained from a single population of each species and from adults rather than seedlings (trait data from seedling life stages was not available at the time of this study), which fails to capture potentially important trait variation across individuals, life stages and populations (Dayrell et al., 2018; Havrilla et al., 2021; Palow et al., 2012; Umaña & Swenson, 2019).

2.3 | Environmental data

We characterized annual drought at each site using DSS, defined as the most extreme cumulative rainfall deficit of evapotranspiration exceeding precipitation reached during the annual dry season, with lower values indicating more severe dry seasons (Condit et al., 2013). Long-term DSS estimates (1961–1990 average) at each site were obtained from Browne et al. (2021). Soil phosphorus levels were obtained from Condit et al. (2013), where soil resin phosphorus levels were estimated using anion-exchange membranes placed in the upper 10 cm of the soil profile during the wet season. We log transformed soil phosphorus levels prior to analysis. While we acknowledge that light available is a major drive of understory dynamics in tropical forests, we were not able to quantify variation in light availability across sites in this study.

2.4 | Growth and mortality model formulation

To estimate the relationships between functional traits and growth and mortality rates for the seedling communities in this study, we fit separate hierarchical Bayesian models for growth and mortality that had similar structures in terms of predictor variables and random effects. We quantified growth using relative growth rates (RGR):

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$$\mathrm{RGR} = \frac{\ln(\mathrm{Height}_2) - \ln(\mathrm{Height}_1)}{(t_2 - t_1)},$$

where t_2 = time two, t_1 = time one, Height_2 = height at time 2, Height_1 = height at time 1. We used a Box–Cox transformation (lambda = 0.15) to reduce skewness and normalize the distribution of growth rates (Condit et al., 2017). This reduced the number of model divergences as well. RGR values were back-transformed to the original scale for presentation in all figures. While our choice of growth metric (relative growth rate of stem height) provides valuable information on the vertical position of the stem in the understory, which influences access to light and competitive dynamics, it fails to account for other types of growth such radial growth, biomass increase or below-ground growth. To reduce noise in growth measurements, we focused only on positive growth rates in this study, although including negative growth rates (due to stem breakage or measurement error) produced qualitatively similar results (results not shown).

In growth models, the response variable (RGR) was assumed to be Normally distributed for each individual observation *i*:

$$\mathrm{RGR}_i \sim \mathrm{Normal}(\hat{y}_i, \sigma).$$

In mortality models, the response variable (1 = dead, 0 = alive) was assumed to be Bernoulli distributed and adjusted to account for varying census intervals (time):

$$\mathrm{Mortality}_i \sim \mathrm{Bernoulli}\left(\mathrm{logit}(\hat{y}_i)^{\mathrm{time}}\right).$$

We estimated overall (i.e. across sites) trait–rate relationships using a model (Model 1) where trait–rate relationships were not allowed to vary across sites:

Model 1: Trait-rates fixed across sites

$$\hat{y}_i \sim \alpha 0 + \alpha 1_{\mathrm{spp}} + \alpha 2_{c,s} + \alpha 3_p + \beta 1 \times \mathrm{InitialHeight}_i + \beta 2 \times \mathrm{Trait}_i,$$

$$\alpha 1_{\mathrm{spp}} \sim \mathrm{Normal}(0, \sigma^2),$$

$$\alpha 2_{c,s} \sim \mathrm{Normal}(0, \sigma^2),$$

$$\alpha 3_p \sim \mathrm{Normal}(0, \sigma^2),$$

where $\hat{y}_i$ is either Box–Cox-transformed RGR or mortality status (1 = dead, 0 = alive) for observation i, $\alpha 0$ is the overall intercept, $\alpha 1_{\mathrm{spp}}$ is a species-level (*spp*) random intercept, $\alpha 2_{c,s}$ is a random effect predicted separately for each census-site combination, $\alpha 3_p$ is a plot-level random effect for each 1 × 1 m seedling plot *p*, $\beta 1$ estimates the effect of height at the previous census on either RGR or mortality and $\beta 2$ estimates each overall trait–rate relationship across all sites. Trait data for LMA, leaf dry matter content, seed mass and leaf area were log-transformed prior to standardization. We then standardized all trait data such that mean = 0 and standard deviation = 1. Original mean and standard deviation values of traits are available in Table S1. We log-transformed and standardized initial seedling height within each species such that mean = 0 and standard deviation = 1 to account for differences in mean seedling height across species.

To estimate how trait–rate relationships varied across sites, we fit an additional model (Model 2) where $\beta 2$ was estimated separately for each site *s*:

Model 2: Trait-rate variable across sites

$$\hat{y}_i \sim \alpha 0 + \alpha 1_{\mathrm{spp}} + \alpha 2_{c,s} + \alpha 3_p + \beta 1 \times \mathrm{InitialHeight}_i + \beta 2_s \times \mathrm{Trait}_i,$$

$$\beta 2_s \sim \mathrm{Normal}(\alpha, \sigma).$$

In this case, $\beta 2_s$ estimates the site-specific slope of a trait on either growth or mortality. To assess whether trait–rate relationships varied substantially across sites, we compared models where $\beta 2$ was allowed to vary across sites (i.e. Model 2) to the model where $\beta 2$ was fixed across sites (i.e. Model 1) using LOOIC (leave-one-out information criterion) (Vehtari et al., 2017). LOOIC is a robust estimate of pointwise out-of-sample prediction accuracy from a fitted Bayesian model based on log-likelihood scores that is on the same scale as other information criteria (e.g. deviance information criterion, Akaike's information criterion—AIC; Vehtari et al., 2017). We considered models within 2 LOOIC units of each other to be equally supported by the data and a model with the lowest LOOIC that was >2 LOOIC units from the next best model to be best fit to the data (following Eisaguirre et al., 2019; Lindenmayer et al., 2022; Ravindran et al., 2021). To calculate LOOIC in this comparison, we used the log-likelihood scores based on RGR_i or $\mathrm{Mortality}_i$, depending on whether it was a growth or mortality model. If the LOOIC score was lower for any model where β_{trait} as allowed to vary across sites compared with the model where $\beta 2$ was fixed across sites, we considered this as evidence that trait–rate relationships varied across sites.

To determine whether variation in trait–rate relationships across sites could be explained by local environmental factors, we used a second-level regression within each mortality and growth model. We linked site-level estimates of trait–rate relationships to DSS (average from 1961 to 1990, lower numbers indicate more severe dry seasons) and soil phosphorus (**SoilP**). We fit three separate model formulations that included either DSS (Model 3) and SoilP (Model 4) as predictors of trait–rate variation across sites ($\beta 2_s$) or an intercept only model with no predictor covariates that allows for variation in trait-rates across sites (Model 2, above) and compared these models using LOOIC, with the log-likelihood calculated based on $\beta 2_s$. Comparing models with DSS or SoilP as predictors to an intercept only model allowed us to test whether these local environmental factors could explain variation in trait–rate relationships across sites. We did not include both DSS and SoilP in a single model because the high collinearity between DSS and SoilP at our sites ($R = -0.74$) and low number of sites ($n = 8$) would lead to reduced power and high uncertainty in parameter estimates when both predictors were used in a single model.

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Model 3: Dry season severity as predictor

$$\beta 2_s \sim \text{Normal}(\alpha + \beta 3 \times \text{DSS}_s, \sigma).$$

Model 4: Soil phosphorus as predictor

$$\beta 2_s \sim \text{Normal}(\alpha + \beta 4 \times \text{Soil}P_s, \sigma).$$

All models were fit using Stan (Carpenter et al., 2017) using the 'RSTAN' package vs. 2.21.2 (Stan Development Team, 2020). Following the Stan prior choice recommendations (Stan Development Team, 2017), we used weakly-informative priors of Half-Normal(0, 1) for variance parameters, $\text{Student}_t(5, 0, 2.5)$ for coefficients in mortality models and $\text{Normal}(0, 1)$ for coefficients in growth models. To ensure no model divergences, which would indicate an issue with model estimation (Stan Development Team, 2020), we used a more informative prior of Half-Normal(0.04, 0.05) for the variance parameter in the second level regression of growth models linking DSS or soil phosphorus to growth-trait variation across sites. Using a weakly-informative prior produced qualitatively similar results. For each model, we ran four independent chains for 3000 iterations, with 1500 iterations of burn in. We checked chain convergence visually and by ensuring the potential scale reduction factor statistic ('rhat') was <1.10 (Kéry, 2010). We calculated LOOIC scores using the 'LOO' R package v. 2.4.1 (Vehtari et al., 2020).

3 | RESULTS

3.1 | Overall trait–rate relationships across sites

We found overall relationships between traits and growth and mortality rates for most traits considered. Annual relative growth rates decreased most strongly with increasing wood density, LMA and seed mass (Figure 3a). Relative growth rates also decreased with increasing leaf dry matter content, but the strength of the relationship between growth rates, maximum height and leaf area was close to 0 (Figure 3a). Average mortality rates decreased strongly with increasing wood density, increasing LMA and increasing leaf dry matter content and increased strongly with increasing maximum height (Figure 3b). The strength of mortality rate–trait relationships was weaker for leaf area and seed mass (Figure 3b).

3.2 | Variation in trait–rate relationships across sites

Trait-growth and trait–mortality relationships were variable across sites for most trait-rate combinations considered, as indicated by ΔLOOIC scores >−2 units for models that allowed sites to vary in their trait–rate relationships than models where trait–rate relationships were fixed across sites (Table 2). The exceptions were LMA–growth, LMA–mortality and leaf area–mortality relationships, where models with trait–rate relationships variable across sites were all within 2 LOOIC units of models with trait–rate relationships fixed across sites (Table 2).

3.3 | Dry season severity and soil phosphorus predicting variation in trait–rate relationships across sites

Variation in trait–growth or trait–mortality relationships was not predicted by DSS in any of the 12 comparisons made, but variation was predicted by soil phosphorus in 2/12 comparisons (Figures 4 and 5), as indicated by ΔLOOIC scores within 2 units for models with

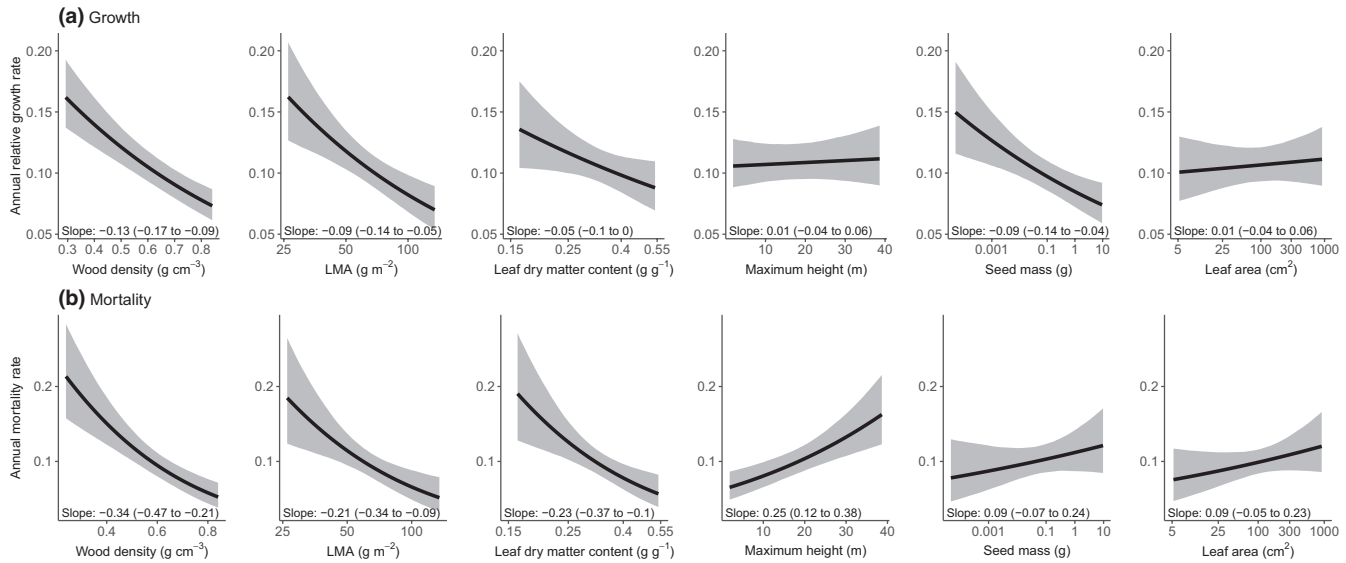


FIGURE 3 Across site trait–rate relationships for growth (a, top row) and mortality (b, bottom row). The black lines show the mean posterior slope estimate, and the shaded regions indicate the 95% credible interval (CI), with the slope estimates and 95% CI also provided as a text inset.

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TABLE 2 Comparisons of models where trait–rate relationships were either held constant across sites ('not variable across sites') or allowed to vary across sites ('variable across sites') for functional traits and separate growth and mortality models. LOOIC is the leave-one-out information criterion, and lower values indicate better model fit. ΔLOOIC is the difference in LOOIC values between the model where trait–rate relationships varied across sites compared with a model where trait–rate relationships were fixed across sites. Values in bold indicate ΔLOOIC values <−2, where the model with variable trait–rate relationships was the better fit.

Vital rate	Trait	Not variable across sites LOOIC (model 1)	Variable across sites LOOIC (model 2)	ΔLOOIC
Growth	Wood density	**26,779.84**	**26,749.18**	**−30.66**
	LMA	17,644.97	17,644.65	−0.32
	Leaf dry matter content	**17,684.52**	**17,663.03**	**−21.49**
	Max height	**17,739.11**	**17,730.37**	**−8.74**
	Seed mass	**23,333.84**	**23,327.05**	**−6.79**
	Leaf area	**17,684.58**	**17,673.17**	**−11.41**
Mortality	Wood density	**14,099.40**	**14,092.81**	**−6.59**
	LMA	7927.00	7927.01	0.01
	Leaf dry matter content	**7965.31**	**7953.73**	**−11.58**
	Max height	**8028.38**	**8021.09**	**−7.29**
	Seed mass	**12,746.35**	**12,737.86**	**−8.49**
	Leaf area	7971.50	7970.46	−1.04

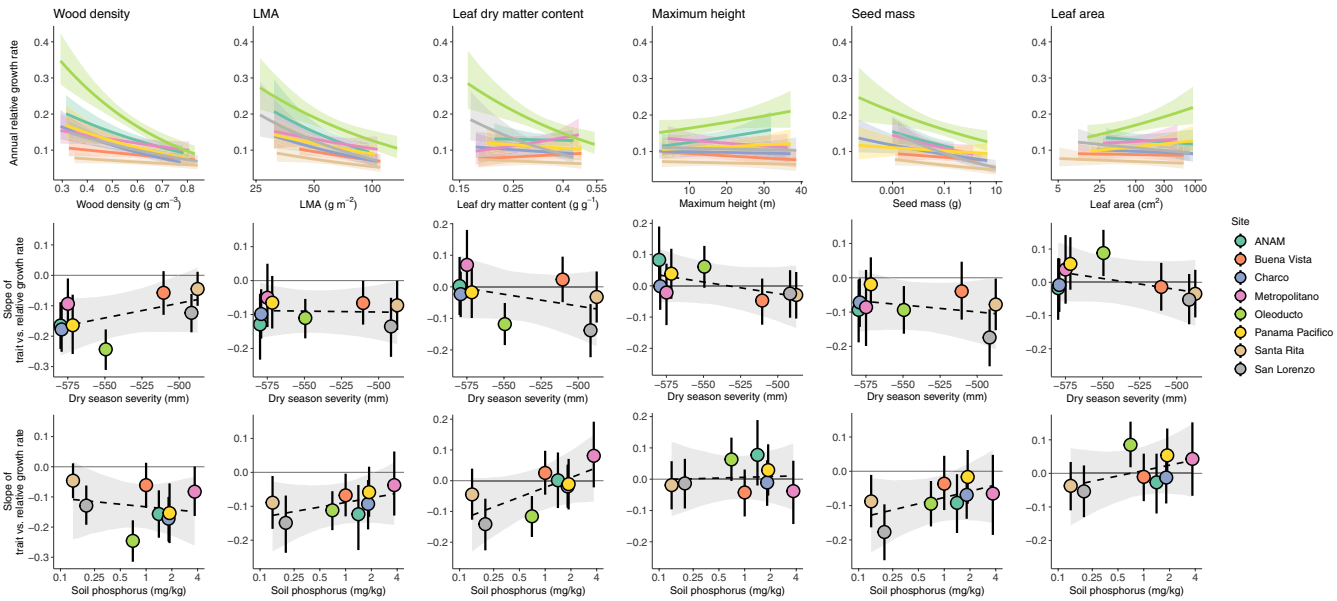


FIGURE 4 Variation of trait–growth rate relationships at eight sites with different environmental conditions. The top row shows for each functional trait the relationship to growth rates for each site (colours with the 95% credible interval shaded). The second and third rows show the mean trait-growth slope for each site (circles, with 95% credible interval) plotted against site-level long-term dry season severity (1961–1990 average) and soil phosphorus (log transformed). If the intercept only model was the best fit model compared with models with either dry season severity or soil phosphorus as covariates, the estimated slope is shown as a dashed line with the 95% credible interval in light grey. The light grey horizontal line shows where the trait–growth relationship is equal to 0.

soil phosphorus as a predictor vs. intercept-only models with no environmental predictors (Table 3).

Variation in trait–growth relationships for wood density, leaf dry matter content, LMA, maximum height, seed mass and leaf area was not explained by variation in either DSS or soil phosphorus across sites (Figure 4, Table 3). In general, there was large uncertainty in these estimates due to the limited number of sites (Table S2).

Wood density–mortality relationships were more strongly negative (higher wood density associated with lower mortality) and maximum height–mortality relationships were more strongly positive (with higher maximum height associated with higher mortality

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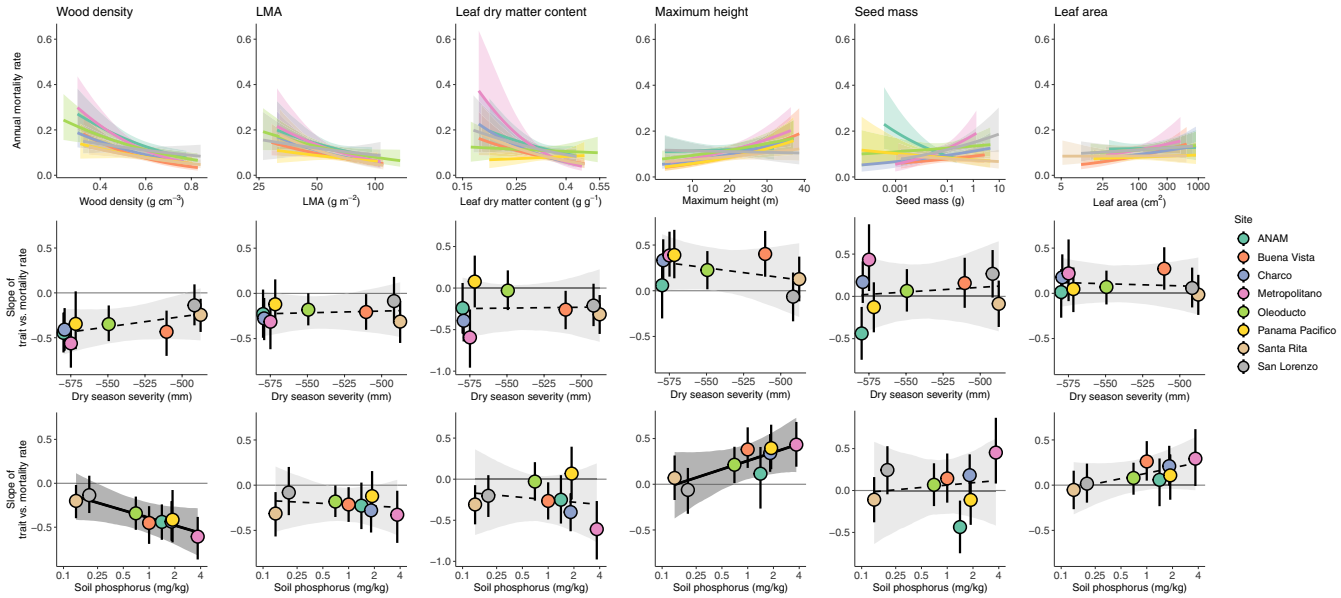


FIGURE 5 Variation of trait–mortality rate relationships at eight individual sites with different environmental conditions. The top row shows for each functional trait the relationship to mortality rates for each site (colours with the 95% credible interval shaded). The second and third rows show the mean trait-mortality slope for each site (circles, with 95% credible interval) plotted against site-level long-term dry season severity (1961–1990 average) and soil phosphorus (log transformed). If models with either dry season severity or soil phosphorus was the best fit model, the estimated slope is shown as a filled black line, and the corresponding 95% credible interval is filled as dark grey. Otherwise, the estimated slope is shown as a dashed line with the 95% credible interval in light grey. The light grey horizontal line shows where the trait–mortality relationship is equal to 0.

TABLE 3 Comparisons of models of how dry season severity (model 3) and soil phosphorus (model 4) explain variation in trait–rate relationships for functional traits and growth and mortality models. The 'dry season severity' and 'soil phosphorus' models contain each environmental factor as a single covariate, while the intercept only model contains no environmental covariates (model 2). LOOIC is the leave-one-out information criterion, and lower values indicate better model fit. ΔLOOIC is the difference in LOOIC values between each model and the intercept only model. Values in bold indicate ΔLOOIC values <−2 where using dry season severity or soil phosphorus as a covariate improved model fit compared with the intercept only model.

Vital rate	Trait	Intercept only LOOIC	Dry season severity LOOIC	Dry season severity ΔLOOIC	Soil phosphorus LOOIC	Soil phosphorus ΔLOOIC
Growth	Wood density	−15.38	−15.85	−0.47	−13.23	2.15
	LMA	−23.30	−20.50	2.80	−21.35	1.95
	Leaf dry matter content	−14.65	−13.47	1.18	−15.65	−1.00
	Max height	−19.22	−18.76	0.46	−16.90	2.32
	Seed mass	−18.86	−17.32	1.54	−18.48	0.38
	Leaf area	−19.62	−18.74	0.88	−18.14	1.48
Mortality	Wood density	−0.02	−1.05	−1.03	**−6.80**	**−6.78**
	LMA	−4.56	−1.07	3.49	0.69	5.25
	Leaf dry matter content	4.76	7.46	2.70	8.00	3.24
	Max height	2.84	4.48	1.64	**−0.02**	**−2.86**
	Seed mass	8.01	10.58	2.57	10.81	2.80
	Leaf area	−1.76	0.74	2.5	−3.02	−1.26

rates) in sites with higher soil phosphorus levels, but neither showed a relationship with DSS (Figure 5). Neither DSS nor soil phosphorus explained variation in trait–mortality relationships for LMA, leaf dry matter content, seed mass or leaf area (Figure 5, Table 3).

4 | DISCUSSION

The relationships between functional traits and vital rates in plants are often weak or inconsistent, and the underlying drivers

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of this pattern are currently unresolved (Paine et al., 2015). In this study, we found widespread evidence that trait–rate relationships varied for seedlings across eight sites along the Isthmus of Panama for a set of common functional traits related to major axes of life-history variation. We found limited support for the hypothesis that local resource availability drives variation in trait–rate relationships. Environmental variables related to water and soil nutrient availability failed to explain site-to-site variation in trait–rate relationships for the majority of cases. In a small subset of cases, soil phosphorus levels explained variation in trait–mortality relationships. These results demonstrate that site-level factors modulate the interaction between functional trait strategies and demographic outcomes, which may explain why trait–rate relationships are inconsistent and weak at broad scales when these factors are not considered.

4.1 | Overall trait–rate relationships across sites

The first goal of our study was to determine whether general trait–rate relationships existed for seedlings across the eight study sites included in this study. Consistent with previous studies in tropical forests (Poorter et al., 2008; Wright et al., 2010), we found general relationships in the expected direction for most trait–rate relationships we considered in this study, despite using trait data collected from adults. We found that species with high wood density, LMA and leaf dry matter content showed lower growth rates and lower mortality rates, following the well-established leaf and wood economics spectrums (Chave et al., 2009; Wright et al., 2004). For seed mass, we found that larger seeds were generally associated with decreased growth rates but contrary to our expectations, were not strongly associated with mortality rates, with the 95% credible interval including 0. The lack of a strong relationship between seed mass and seedling mortality is partly in contrast to previous studies that have found increased seed mass is related to increased seedling establishment and survival; however, this relationship can decouple as seedlings age and rely less on the resources provided by the seed (Dalling & Hubbell, 2002; Moles & Westoby, 2006; Westoby et al., 2002). The 200 mm minimum height cutoff used in this study encompasses a range of seedling ages, most of which may be beyond the point of relying on the seed for sustenance, which could explain the absence of a negative relationship between seed mass and mortality rates. We found that higher trait values of maximum height were associated with increased mortality rates but not strongly associated with growth rates, while leaf area was not strongly associated with either growth or mortality rates. These results are partially consistent with previous studies that have found that species with high maximum height ('long-lived pioneers') tend to have lower seedling performance (Rüger et al., 2018). The lack of a general relationship for maximum height, leaf area and growth rates could be explained by wide site-to-site variation in these trait–rate relationships across our eight sites (see below).

4.2 | Variation in trait–rate relationships across sites

Our second question focused on whether the strength of trait–rate relationships varied substantially among our study sites. We found strong support for the hypothesis that trait–rate relationships vary among sites, with all functional traits analysed except for LMA-growth, LMA-mortality and leaf area–mortality relationships showing evidence of variation across sites. While a goal of this study was to test whether resource availability could explain variation in trait–rate relationships across sites, differences in species composition could also lead to variation in trait–rate relationships, especially if compositional differences across sites are due to species turnover rather than nestedness. In our dataset, the majority of species (59%) occurred in at least two sites and our statistical models included a species-level random intercept that in theory would help control for differences in species composition across sites, especially in cases where species turnover is not complete. However, in our study area, there is a significant level of species turnover across sites related to differences in both precipitation and soil phosphorus (Pyke et al., 2001; Umaña et al., 2021), which may contribute to the variation in functional trait–rate relationships across sites we observed.

In some cases, such as wood density–mortality relationships, the slope estimates remained consistently in the same direction across all sites, although the magnitude of the slope estimate varied on a site-by-site basis. In contrast, for maximum height–growth, leaf area–growth and leaf dry matter content–growth relationships, slope estimates varied between positive, negative or close to 0 across sites, obscuring a general pattern in trait–rate relationships for these traits. Taken together, these results suggest that at least for some traits, a single-site study is unlikely to fully capture the potential variability or even accurately estimate the general direction of a trait–rate relationship. Therefore, care must be taken when extrapolating the results from single sites to other communities in different environmental contexts. Additionally, a general relationship between functional traits and a vital rate does not preclude the existence of considerable site-by-site variation that may be explained by site-level factors. Conversely, the lack of a general trait–rate relationship may arise due to wide site-to-site variation and thus weak overall trait–rate relationships do not imply that a given trait is not relevant for vital rates at a particular site.

4.3 | Dry season severity and soil phosphorus predicting variation in trait–rate relationships across sites

For our third question, we tested the hypothesis that variation in local resource availability explains variation in trait–rate relationships across sites, with the expectation that lower-resource environments would lead to weaker trait–growth relationships and stronger trait–mortality relationships (Figure 1). We found limited support for

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this hypothesis. Overall, we found limited statistical support that models including DSS or soil phosphorous performed better than a more parsimonious model that did not use environmental predictors to explain variation in trait–rate relationships across sites. Altogether, none of the trait–growth and trait–mortality relationships were predicted by DSS or and 2/6 trait–mortality relationships were predicted by soil phosphorus, and neither of those cases were in the predicted direction (Table 1).

Contrary to our expectations that trait–mortality relationships would be strongest at sites with low soil phosphorus (Figure 1b), we found more strongly negative wood density–mortality relationships and more strongly positive maximum height–mortality relationships in sites with high soil phosphorus, indicating that species with low wood density and higher maximum height suffered relatively high mortality at high phosphorus sites compared with low phosphorus sites. A potential explanation for these unexpected results is that herbivory tends to be higher at sites with high levels of soil phosphorus, which also tend to be drier sites (Muehleisen et al., 2020; Weissflog et al., 2018). Furthermore, experimental studies have shown that phosphorus addition increases herbivory pressure on seedlings, implicating phosphorus as a causal mechanism driving herbivory (Santiago et al., 2012). Taking this into account, any potential benefits to competitive ability of low wood density or high maximum height at sites with high soil phosphorus may be offset by higher rates of herbivory, suggesting that herbivory may be a limiting factor for vital rates of seedlings at these sites, although future experimental work is needed to confirm this hypothesis.

5 | CONCLUSIONS

In summary, we showed through a long-term demographic study of thousands of seedlings across the Isthmus of Panama that there is widespread variation in trait–rate relationships across sites, that some of this variation was explained by site-level differences in soil phosphorus, but mostly remains unexplained. Future studies that assess a wider range of environmental covariates, including variation in light and ecological processes like herbivory, along with considering variation in traits across individuals, life stages and populations, will likely improve the amount of variation in trait–rate relationships explained across environmental gradients. More broadly, accounting for site-to-site variation and acknowledging the context-dependent nature of trait–rate relationships may help resolve a fundamental issue in trait-based ecology that many studies show weak to non-existent relationships between functional traits and vital rates.

AUTHOR CONTRIBUTIONS

Liza S. Comita and Luke Browne conceived the study. Robert Bagchi, S. Joseph Wright and Lars Markesteijn contributed to study design and establishment. Lars Markesteijn, Liza S. Comita, Eric Manzané-Pinzón and Luke Browne participated in data collection. Luke Browne performed data analysis with input from Liza S. Comita. Luke Browne wrote the first draft of the manuscript, and all authors contributed to the interpretation and writing. All authors approve of the submitted version of the manuscript.

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CONFLICT OF INTEREST

The authors declare no competing interests.

DATA AVAILABILITY STATEMENT

Data and model code deposited in the Dryad Digital Repository http://doi.org/10.5061/dryad.mkkwh713s (Browne et al., 2022).

ORCID

Luke Browne https://orcid.org/0000-0002-2239-3007
Lars Markesteijn https://orcid.org/0000-0003-3046-3121
S. Joseph Wright https://orcid.org/0000-0003-4260-5676
Robert Bagchi https://orcid.org/0000-0003-4035-4105
Liza S. Comita https://orcid.org/0000-0002-9169-1331

REFERENCES

Ackerly, D. D. (2003). Community assembly, niche conservatism, and adaptive evolution in changing environments. *International Journal of Plant Sciences*, *164*(S3), S165–S184. https://doi.org/10.1086/368401

Alvarez-Clare, S., & Kitajima, K. (2007). Physical defence traits enhance seedling survival of neotropical tree species. *Functional Ecology*, *21*(6), 1044–1054. https://doi.org/10.1111/j.1365-2435.2007.01320.x

Alvarez-Clare, S., Mack, M. C., & Brooks, M. (2013). A direct test of nitrogen and phosphorus limitation to net primary productivity in a lowland tropical wet forest. *Ecology*, *94*(7), 1540–1551. https://doi.org/10.1890/12-2128.1

Bloor, J. M. G., & Grubb, P. J. (2003). Growth and mortality in high and low light: Trends among 15 shade-tolerant tropical rain forest tree species. *Journal of Ecology*, *91*(1), 77–85. https://doi.org/10.1046/j.1365-2745.2003.00743.x

Browne, L., Markesteijn, L., Engelbrecht, B. M. J., Jones, F. A., Lewis, O. T., Manzané-Pinzón, E., Wright, S. J., & Comita, L. S. (2021). Increased mortality of tropical tree seedlings during the extreme 2015-16 El Niño. *Global Change Biology*, *27*(20), 5043–5053. https://doi.org/10.1111/gcb.15809

Browne, L., Markesteijn, L., Manzané-Pinzón, E., Wright, S. J., Bagchi, R., Engelbrecht, B., Jones, F. A., & Comita, L. S. (2022). Widespread

13652435, 0, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.14213 by Yale University, Wiley Online Library on [11/11/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

variation in functional trait-vital rate relationships in tropical tree seedlings across a precipitation and soil phosphorus gradient. *Dryad Digital Repository*, https://doi.org/10.5061/dryad.mkkwh713s

Cadotte, M. W., Carscadden, K., & Mirotchnick, N. (2011). Beyond species: Functional diversity and the maintenance of ecological processes and services. *Journal of Applied Ecology*, *48*(5), 1079–1087. https://doi.org/10.1111/j.1365-2664.2011.02048.x

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of Statistical Software*, *76*(1), 1–32. https://doi.org/10.18637/jss.v076.i01

Chave, J., Coomes, D., Jansen, S., Lewis, S. L., Swenson, N. G., & Zanne, A. E. (2009). Towards a worldwide wood economics spectrum. *Ecology Letters*, *12*(4), 351–366. https://doi.org/10.1111/j.1461-0248.2009.01285.x

Coley, P. D., Bryant, J. P., & Chapin, F. S., 3rd. (1985). Resource availability and plant antiherbivore defense. *Science*, *230*(4728), 895–899. https://doi.org/10.1126/science.230.4728.895

Comita, L. S., Aguilar, S., Pérez, R., Lao, S., & Hubbell, S. P. (2007). Patterns of woody plant species abundance and diversity in the seedling layer of a tropical forest. *Journal of Vegetation Science*, *18*(2), 163. https://doi.org/10.1658/1100-9233(2007)18[163:powpsa]2.0.co;2

Condit, R., Engelbrecht, B. M. J., Pino, D., Pérez, R., & Turner, B. L. (2013). Species distributions in response to individual soil nutrients and seasonal drought across a community of tropical trees. *Proceedings of the National Academy of Sciences of the United States of America*, *110*(13), 5064–5068. https://doi.org/10.1073/pnas.1218042110

Condit, R., Pérez, R., Lao, S., Aguilar, S., & Hubbell, S. P. (2017). Demographic trends and climate over 35 years in the Barro Colorado 50 ha plot. *Forest Ecosystems*, *4*(1), 17. https://doi.org/10.1186/s40663-017-0103-1

Dalling, J. W., & Hubbell, S. P. (2002). Seed size, growth rate and gap microsite conditions as determinants of recruitment success for pioneer species. *Journal of Ecology*, *90*(3), 557–568. https://doi.org/10.1046/j.1365-2745.2002.00695.x

Dayrell, R. L. C., Arruda, A. J., Pierce, S., Negreiros, D., Meyer, P. B., Lambers, H., & Silveira, F. A. O. (2018). Ontogenetic shifts in plant ecological strategies. *Functional Ecology*, *32*(12), 2730–2741. https://doi.org/10.1111/1365-2435.13221

Díaz, S., Kattge, J., Cornelissen, J. H. C., Wright, I. J., Lavorel, S., Dray, S., Reu, B., Kleyer, M., Wirth, C., Prentice, I. C., Garnier, E., Bönisch, G., Westoby, M., Poorter, H., Reich, P. B., Moles, A. T., Dickie, J., Gillison, A. N., Zanne, A. E., … Gorné, L. D. (2016). The global spectrum of plant form and function. *Nature*, *529*(7585), 167–171. https://doi.org/10.1038/nature16489

Eisaguirre, J. M., Auger-Méthé, M., Barger, C. P., Lewis, S. B., Booms, T. L., & Breed, G. A. (2019). Dynamic-parameter movement models reveal drivers of migratory pace in a soaring bird. *Frontiers in Ecology and Evolution*, *7*. https://doi.org/10.3389/fevo.2019.00317

Gaviria, J., Turner, B. L., & Engelbrecht, B. M. J. (2017). Drivers of tree species distribution across a tropical rainfall gradient. *Ecosphere*, *8*(2), e01712. https://doi.org/10.1002/ecs2.1712

Green, P. T., Harms, K. E., & Connell, J. H. (2014). Nonrandom, diversifying processes are disproportionately strong in the smallest size classes of a tropical forest. *Proceedings of the National Academy of Sciences of the United States of America*, *111*(52), 18649–18654. https://doi.org/10.1073/pnas.1321892112

Grime, J. P. (1977). Evidence for the existence of three primary strategies in plants and its relevance to ecological and evolutionary theory. *The American Naturalist*, *111*(982), 1169–1194. https://doi.org/10.1086/283244

Havrilla, C. A., Munson, S. M., Yackulic, E. O., & Butterfield, B. J. (2021). Ontogenetic trait shifts: Seedlings display high trait variability during early stages of development. *Functional Ecology*, *35*(11), 2409–2423. https://doi.org/10.1111/1365-2435.13897

Jiang, F., & Jin, G. (2021). Functional traits influence plant survival depending on environmental contexts and life stages in an old-growth temperate forest. *Journal of Plant Ecology*, *14*(5), 981–994. https://doi.org/10.1093/jpe/rtab049

Kéry, M. (2010). *Introduction to WinBUGS for ecologists*. Academic Press. https://doi.org/10.1016/B978-0-12-378605-0.00022-3

King, D. A., Wright, S. J., & Connell, J. H. (2006). The distribution of maximum tree heights in tropical vs. temperate forests: Cause or consequence of tropical diversity. *Journal of Tropical Ecology*, *22*, 11–24.

Kobe, R. K. (1999). Light gradient partitioning among tropical tree species through differential seedling mortality and growth. *Ecology*, *80*(1), 187–201. https://doi.org/10.1890/0012-9658(1999)080[0187:lgpatt]2.0.co;2

Kohyama, T., Suzuki, E., Partomihardjo, T., Yamada, T., & Kubo, T. (2003). Tree species differentiation in growth, recruitment and allometry in relation to maximum height in a Bornean mixed dipterocarp forest. *Journal of Ecology*, *91*(5), 797–806. https://doi.org/10.1046/j.1365-2745.2003.00810.x

Kraft, N. J. B., Adler, P. B., Godoy, O., James, E. C., Fuller, S., & Levine, J. M. (2015). Community assembly, coexistence and the environmental filtering metaphor. *Functional Ecology*, *29*(5), 592–599. https://doi.org/10.1111/1365-2435.12345

Laughlin, D. C. (2014). The intrinsic dimensionality of plant traits and its relevance to community assembly. *Journal of Ecology*, *102*(1), 186–193. https://doi.org/10.1111/1365-2745.12187

Laughlin, D. C. (2018). Rugged fitness landscapes and Darwinian demons in trait-based ecology. *The New Phytologist*, *217*(2), 501–503. https://doi.org/10.1111/nph.14908

Laughlin, D. C., & Messier, J. (2015). Fitness of multidimensional phenotypes in dynamic adaptive landscapes. *Trends in Ecology & Evolution*, *30*(8), 487–496. https://doi.org/10.1016/j.tree.2015.06.003

Laughlin, D. C., Strahan, R. T., Adler, P. B., & Moore, M. M. (2018). Survival rates indicate that correlations between community-weighted mean traits and environments can be unreliable estimates of the adaptive value of traits. *Ecology Letters*, *21*(3), 411–421. https://doi.org/10.1111/ele.12914

Lavorel, S., & Garnier, E. (2002). Predicting changes in community composition and ecosystem functioning from plant traits: Revisiting the holy grail. *Functional Ecology*, *16*(5), 545–556. https://doi.org/10.1046/j.1365-2435.2002.00664.x

Li, Y., Jiang, Y., Zhao, K., Chen, Y., Wei, W., Shipley, B., & Chu, C. (2022). Exploring trait-performance relationships of tree seedlings along experimentally manipulated light and water gradients. *Ecology*, *103*, e3703. https://doi.org/10.1002/ecy.3703

Lindenmayer, D., Blanchard, W., Bowd, E., Scheel, B., Tyrone, F. C. L., & David, M. L. B. (2022). Rapid bird species recovery following high-severity wildfire but in the absence of early successional specialists. *Diversity and Discrimination*, *28*, 2110–2123. https://doi.org/10.1111/ddi.13611

McGill, B. J., Enquist, B. J., Weiher, E., & Westoby, M. (2006). Rebuilding community ecology from functional traits. *Trends in Ecology & Evolution*, *21*(4), 178–185. https://doi.org/10.1016/j.tree.2006.02.002

Moles, A. T., & Westoby, M. (2006). Seed size and plant strategy across the whole life cycle. *Oikos*, *113*(1), 91–105. https://doi.org/10.1111/j.0030-1299.2006.14194.x

Muehleisen, A. J., Engelbrecht, B. M. J., Jones, F. A., Manzané-Pinzón, E., & Comita, L. S. (2020). Local adaptation to herbivory within tropical tree species along a rainfall gradient. *Ecology*, *101*(11), e03151. https://doi.org/10.1002/ecy.3151

Muller-Landau, H. C. (2010). The tolerance-fecundity trade-off and the maintenance of diversity in seed size. *Proceedings of the National Academy of Sciences of the United States of America*, *107*(9), 4242–4247. https://doi.org/10.1073/pnas.0911637107

13652435, 0, Downloaded from https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.14213 by Yale University, Wiley Online Library on [11/11/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Paine, C. E. T., Amissah, L., Auge, H., Baraloto, C., Baruffol, M., Bourland, N., Bruelheide, H., Daïnou, K., de Gouvenain, R. C., Doucet, J.-L., Doust, S., Fine, P. V. A., Fortunel, C., Haase, J., Holl, K. D., Jactel, H., Li, X., Kitajima, K., Koricheva, J., … Hector, A. (2015). Globally, functional traits are weak predictors of juvenile tree growth, and we do not know why. *Journal of Ecology*, *103*(4), 978–989. https://doi.org/10.1111/1365-2745.12401

Palow, D. T., Nolting, K., & Kitajima, K. (2012). Functional trait divergence of juveniles and adults of nine Inga species with contrasting soil preference in a tropical rain forest. *Functional Ecology*, *26*(5), 1144–1152. https://doi.org/10.1111/j.1365-2435.2012.02019.x

Poorter, L. (2007). Are species adapted to their regeneration niche, adult niche, or both? *The American Naturalist*, *169*(4), 433–442. https://doi.org/10.1086/512045

Poorter, L. (2009). Leaf traits show different relationships with shade tolerance in moist versus dry tropical forests. *The New Phytologist*, *181*(4), 890–900. https://doi.org/10.1111/j.1469-8137.2008.02715.x

Poorter, L., Castilho, C. V., Schietti, J., Oliveira, R. S., & Costa, F. R. C. (2018). Can traits predict individual growth performance? A test in a hyperdiverse tropical forest. *The New Phytologist*, *219*(1), 109–121. https://doi.org/10.1111/nph.15206

Poorter, L., & Rozendaal, D. M. A. (2008). Leaf size and leaf display of thirty-eight tropical tree species. *Oecologia*, *158*(1), 35–46. https://doi.org/10.1007/s00442-008-1131-x

Poorter, L., Wright, S. J., Paz, H., Ackerly, D. D., Condit, R., Ibarra-Manríquez, G., Harms, K. E., Licona, J. C., Martínez-Ramos, M., Mazer, S. J., Muller-Landau, H. C., Peña-Claros, M., Webb, C. O., & Wright, I. J. (2008). Are functional traits good predictors of demographic rates? Evidence from five neotropical forests. *Ecology*, *89*(7), 1908–1920. https://doi.org/10.1890/07-0207.1

Pyke, C. R., Condit, R., Aguilar, S., & Lao, S. (2001). Floristic composition across a climatic gradient in a neotropical lowland forest. *Journal of Vegetation Science: Official Organ of the International Association for Vegetation Science*, *12*(4), 553–566. https://doi.org/10.2307/3237007

Ravindran, S., Froy, H., Underwood, S. L., Dorrens, J., Seeker, L. A., Watt, K., Wilbourn, R. V., Pilkington, J. G., Harrington, L., Pemberton, J. M., & Nussey, D. H. (2021). The association between female reproductive performance and leukocyte telomere length in wild Soay sheep. *Molecular Ecology*. https://doi.org/10.1111/mec.16175

Reich, P. B. (2014). The world-wide 'fast–slow' plant economics spectrum: A traits manifesto. *Journal of Ecology*, *102*(2), 275–301. https://doi.org/10.1111/1365-2745.12211

Rüger, N., Comita, L. S., Condit, R., Purves, D., Rosenbaum, B., Visser, M. D., Wright, S. J., & Wirth, C. (2018). Beyond the fast-slow continuum: Demographic dimensions structuring a tropical tree community. *Ecology Letters*, *21*(7), 1075–1084. https://doi.org/10.1111/ele.12974

Rüger, N., Condit, R., Dent, D. H., DeWalt, S. J., Hubbell, S. P., Lichstein, J. W., Lopez, O. R., Wirth, C., & Farrior, C. E. (2020). Demographic trade-offs predict tropical forest dynamics. *Science*, *368*(6487), 165–168. https://doi.org/10.1126/science.aaz4797

Russo, S. E., Brown, P., Tan, S., & Davies, S. J. (2007). Interspecific demographic trade-offs and soil-related habitat associations of tree species along resource gradients: Demographic trade-offs on habitat gradients. *The Journal of Ecology*, *96*(1), 192–203. https://doi.org/10.1111/j.1365-2745.2007.01330.x

Russo, S. E., Davies, S. J., King, D. A., & Tan, S. (2005). Soil-related performance variation and distributions of tree species in a Bornean rain forest. *Journal of Ecology*, *93*(5), 879–889. https://doi.org/10.1111/j.1365-2745.2005.01030.x

Santiago, L. S., Wright, S. J., Harms, K. E., Yavitt, J. B., Korine, C., Garcia, M. N., & Turner, B. L. (2012). Tropical tree seedling growth responses to nitrogen, phosphorus and potassium addition. *Journal of Ecology*, *100*(2), 309–316. https://doi.org/10.1111/j.1365-2745.2011.01904.x

Stan Development Team. (2017). *Prior choice recommendations*. Stan Wiki. https://github.com/stan-dev/stan/wiki/Prior-Choice-Recommendations

Stan Development Team. (2020). *RStan: The R interface to Stan* (2.21.2) [Computer software]. http://mc-stan.org/

Stearns, S. C. (1992). *The evolution of life histories*. Oxford University Press.

Swenson, N. G., Worthy, S. J., Eubanks, D., Iida, Y., Monks, L., Petprakob, K., Rubio, V. E., Staiger, K., & Zambrano, J. (2020). A reframing of trait–demographic rate analyses for ecology and evolutionary biology. *International Journal of Plant Sciences*, *181*(1), 33–43. https://doi.org/10.1086/706189

Umaña, M. N., Condit, R., Pérez, R., Turner, B. L., Wright, S. J., & Comita, L. S. (2021). Shifts in taxonomic and functional composition of trees along rainfall and phosphorus gradients in Central Panama. *Journal of Ecology*, *109*(1), 51–61. https://doi.org/10.1111/1365-2745.13442

Umaña, M. N., & Swenson, N. G. (2019). Intraspecific variation in traits and tree growth along an elevational gradient in a subtropical forest. *Oecologia*, *191*(1), 153–164. https://doi.org/10.1007/s00442-019-04453-6

Umaña, M. N., Zhang, C., Cao, M., Lin, L., & Swenson, N. G. (2017). A core-transient framework for trait-based community ecology: An example from a tropical tree seedling community. *Ecology Letters*, *20*, 619–628. https://doi.org/10.1111/ELE.12760

Vehtari, A., Gabry, J., Magnusson, M., Yao, Y., Bürkner, P.-C., Paananen, T., & Gelman, A. (2020). *Loo: Efficient leave-one-out cross-validation and WAIC for Bayesian models* (R package version 2.4.1) [computer software]. https://mc-stan.org/loo/

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, *27*(5), 1413–1432. https://doi.org/10.1007/s11222-016-9696-4

Violle, C., Navas, M.-L., Vile, D., Kazakou, E., Fortunel, C., Hummel, I., & Garnier, E. (2007). Let the concept of trait be functional! *Oikos*, *116*(5), 882–892. https://doi.org/10.1111/j.0030-1299.2007.15559.x

Visser, M. D., Bruijning, M., Wright, S. J., Muller-Landau, H. C., Jongejans, E., Comita, L. S., & de Kroon, H. (2016). Functional traits as predictors of vital rates across the life cycle of tropical trees. *Functional Ecology*, *30*(2), 168–180. https://doi.org/10.1111/1365-2435.12621

Walters, M. B., & Reich, P. B. (1996). Are shade tolerance, survival, and growth linked? Low light and nitrogen effects on hardwood seedlings. *Ecology*, *77*(3), 841–853. https://doi.org/10.2307/2265505

Weiher, E., Werf, A., Thompson, K., Roderick, M., Garnier, E., & Eriksson, O. (1999). Challenging Theophrastus: A common core list of plant traits for functional ecology. *Journal of Vegetation Science*, *10*(5), 609–620. https://doi.org/10.2307/3237076

Weissflog, A., Markesteijn, L., Lewis, O. T., Comita, L. S., & Engelbrecht, B. M. J. (2018). Contrasting patterns of insect herbivory and predation pressure across a tropical rainfall gradient. *Biotropica*, *50*(2), 302–311. https://doi.org/10.1111/btp.12513

Westoby, M., Falster, D. S., Moles, A. T., Vesk, P. A., & Wright, I. J. (2002). Plant ecological strategies: Some leading dimensions of variation between species. *Annual Review of Ecology and Systematics*, *33*(1), 125–159. https://doi.org/10.1146/annurev.ecolsys.33.010802.150452

Worthy, S. J., Laughlin, D. C., Zambrano, J., Umaña, M. N., Zhang, C., Lin, L., Cao, M., & Swenson, N. G. (2020). Alternative designs and tropical tree seedling growth performance landscapes. *Ecology*, *101*, e03007. https://doi.org/10.1002/ecy.3007

Worthy, S. J., & Swenson, N. G. (2019). Functional perspectives on tropical tree demography and forest dynamics. *Ecological Processes*, *8*(1), 1. https://doi.org/10.1186/s13717-018-0154-4

Wright, I. J., Reich, P. B., Westoby, M., Ackerly, D. D., Baruch, Z., Bongers, F., Cavender-Bares, J., Chapin, T., Cornelissen, J. H. C., Diemer, M., Flexas, J., Garnier, E., Groom, P. K., Gulias, J., Hikosaka, K., Lamont, B. B., Lee, T., Lee, W., Lusk, C., … Villar, R. (2004). The worldwide leaf economics spectrum. *Nature*, *428*(6985), 821–827. https://doi.org/10.1038/nature02403

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Wright, I. J., & Westoby, M. (2002). Leaves at low versus high rainfall: Coordination of structure, lifespan and physiology. *The New Phytologist*, *155*(3), 403–416. https://doi.org/10.1046/j.1469-8137.2002.00479.x

Wright, S. J., Kitajima, K., Kraft, N. J. B., Reich, P. B., Wright, I. J., Bunker, D. E., Condit, R., Dalling, J. W., Davies, S. J., Díaz, S., Engelbrecht, B. M. J., Harms, K. E., Hubbell, S. P., Marks, C. O., Ruiz-Jaen, M. C., Salvador, C. M., & Zanne, A. E. (2010). Functional traits and the growth-mortality trade-off in tropical trees. *Ecology*, *91*(12), 3664–3674. https://doi.org/10.1890/09-2335.1

Wright, S. J., Yavitt, J. B., Wurzburger, N., Turner, B. L., Tanner, E. V. J., Sayer, E. J., Santiago, L. S., Kaspari, M., Hedin, L. O., Harms, K. E., Garcia, M. N., & Corre, M. D. (2011). Potassium, phosphorus, or nitrogen limit root allocation, tree growth, or litter production in a lowland tropical forest. *Ecology*, *92*(8), 1616–1625. https://doi.org/10.1890/10-1558.1

Yang, J., Cao, M., & Swenson, N. G. (2018). Why functional traits do not predict tree demographic rates. *Trends in Ecology & Evolution*, *33*(5), 326–336. https://doi.org/10.1016/j.tree.2018.03.003

Yang, J., Song, X., Zambrano, J., Chen, Y., Cao, M., Deng, X., Zhang, W., Yang, X., Zhang, G., Tang, Y., & Swenson, N. G. (2021). Intraspecific variation in tree growth responses to neighbourhood composition and seasonal drought in a tropical forest. *The Journal of Ecology*, *109*(1), 26–37. https://doi.org/10.1111/1365-2745.13439

Zalamea, P.-C., Turner, B. L., Winter, K., Jones, F. A., Sarmiento, C., & Dalling, J. W. (2016). Seedling growth responses to phosphorus reflect adult distribution patterns of tropical trees. *The New Phytologist*, *212*(2), 400–408. https://doi.org/10.1111/nph.14045

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